Force, Mass and Motion: From Push to Prediction

PhET Forces and Motion: Basics · PhET

Years 7-10	~30 minutes	Science · Physics

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Simulation: https://phet.colorado.edu/en/simulations/forces-and-motion-basics
Curriculum links: AC9S7U04 · AC9S10U05

Learning intentions

- I can describe how the size of a force changes an object's motion.
- I can explain the effect of friction on a moving object.
- I can use force and acceleration data from the simulation to describe the relationship between force, mass and acceleration.

Getting started

Open the simulation and start on the first screen, where you push crates of different sizes across the ground. Try a few pushes freely first, then work through the tasks below in order. Later tasks ask you to switch to the graph or Acceleration Lab screen to look at the numbers behind what you observed.

Tasks

1. Choose the small crate. Apply a small push and describe what happens to its speed.

2. Now apply a much bigger push to the same crate. What changes about how it moves?

3. Switch the surface to ice (frictionless) and push the crate once. What happens to the crate after you stop pushing? Why do you think this happens?

4. Switch back to a normal (friction) surface. Push each crate size once with the same force, then stop. Record what happens to each.

Crate size	What happens after you stop pushing

5. Based on your table, does a heavier crate need a bigger or smaller push to start moving? Explain using the word 'friction'.

6. Switch to the graph or Acceleration Lab screen. Apply a steady force (e.g. 50 N) to an object and note the acceleration reading, then try a larger force (e.g. 100 N) on the same mass. Calculate the ratio between the two forces and the ratio between the two accelerations. What do you notice?

Working:

7. Keep the applied force the same but change the mass of the object. Describe how the acceleration changes, and connect this to the idea that force, mass and acceleration are related.

8. In your own words, explain the difference between a 'balanced' and an 'unbalanced' force, using an example from this simulation.

Extension (optional)

A car on ice and a shopping trolley on carpet both need a push to start moving. Use what you learned about friction and mass to explain which one would keep moving for longer after you stop pushing, and why.